

Trading Agent Post-Mortem Report

Hypotheses

The following hypotheses were formulated by the multi-panelist forum to guide the investigation of agent trading behaviour.

Hypothesis 1

Hypothesis: PPO exhibits a high-frequency accumulation bias compared to other agents, leading to excessive transaction costs. **Rationale:** PPO shows a significantly higher buy-to-sell ratio and transaction count (1539 buys) compared to DDPG or TD3, suggesting it is overly sensitive to short-term price fluctuations. **Key Factors:**

- Transaction count per agent
- Buy/Sell ratio
- Average holding period

Possible Tests:

- Compare transaction costs against a passive benchmark
- Correlate buy frequency with volatility indicators in the synthetic dataset

Code grounding: Relevant code paths: `envs.gym_wrappers.trading_env.py`, `evaluation.runner.py` Relevant config keys: `ppo_params` Exercised flow: The agent interacts with the `TradingEnv`, where transaction costs are calculated per trade, and the runner evaluates the resulting transaction frequency and cost metrics. Why this code matters: PPO’s high-frequency trading behavior is likely driven by the interaction between its policy update mechanism and the transaction cost penalties defined in the `TradingEnv`. The agent’s tendency to accumulate positions is captured by the transaction cost logic in the environment. Evidence summary: The evidence is indirect; the hypothesis relies on the interaction between PPO’s policy updates and the transaction cost logic defined in the trading environment, which is evaluated by the runner. Supporting paths: `envs.gym_wrappers.trading_env.py`, `evaluation.runner.py` Evidence config keys: `ppo_params` Evidence snippets: PPO’s high-frequency trading behavior is likely driven by the interaction between its policy update mechanism and the transaction cost penalties defined in the `TradingEnv`.

Hypothesis 2

Hypothesis: Agents are suffering from ‘regime-switch paralysis’ due to the interleaved synthetic data structure. **Rationale:** The high number of HOLD actions across all agents (exceeding 85% of total steps) suggests that agents are failing to adapt their strategies when the market regime changes within the interleaved dataset. **Key Factors:**

- Hold action percentage
- Portfolio turnover rate at regime boundaries
- Sharpe ratio degradation at regime transition points

Possible Tests:

- Analyze portfolio turnover specifically during the 5-day window surrounding regime switches
- Compare agent performance on isolated regimes versus interleaved regimes

Code grounding: Relevant code paths: `data.fetchers.experiment_data_loader.py` Relevant config keys: `data_source` Exercised flow: The data loader reads synthetic CSV files, which are then fed into the environment; the agent’s inability to adapt to these shifts is observed during the evaluation phase. Why this code matters: The synthetic data structure, when interleaved, may cause agents to struggle with identifying regime boundaries, leading to inaction or ‘paralysis’ during transitions. Evidence summary: The evidence is indirect; the data loader handles synthetic CSV files, and the hypothesis suggests that the interleaved structure obscures regime boundaries, causing agent inaction. Supporting paths: `data.fetchers.experiment_data_loader.py` Evidence config keys: `data_source` Evidence snippets: The synthetic data structure, when interleaved, may cause agents to struggle with identifying regime boundaries, leading to inaction or ‘paralysis’ during transitions.

Hypothesis 3

Hypothesis: A2C and DDPG are prone to ‘ticker-specific concentration risk’ due to limited exploration during training. **Rationale:** The peak allocations show heavy concentration in specific tickers (e.g., PPO’s 45% allocation in DIS), indicating the agents are not effectively diversifying their portfolios. **Key Factors:**

- Herfindahl-Hirschman Index (HHI) of portfolio weights
- Peak allocation percentage per ticker
- Number of unique tickers traded

Possible Tests:

- Calculate the HHI for each agent’s portfolio over time
- Test if limiting max weight per ticker in the environment improves risk-adjusted returns

Code grounding: Relevant code paths: `evaluation.experiment_setup.py` Relevant config keys: `a2c_params`, `ddpg_params` Exercised flow: The agent is initialized with specific hyperparameters in `experiment_setup.py`, and the training process fails to explore the full ticker space. Why this code matters: Limited exploration in A2C and DDPG, potentially due to insufficient noise or entropy coefficients, leads to the agents focusing on a narrow subset of tickers. Evidence

summary: The evidence is indirect; the hypothesis links initialization parameters in the experiment setup to a failure in exploring the full ticker space, resulting in concentration. Supporting paths: `evaluation.experiment_setup.py` Evidence config keys: `a2c_params`, `ddpg_params` Evidence snippets: Limited exploration in A2C and DDPG, potentially due to insufficient noise or entropy coefficients, leads to the agents focusing on a narrow subset of tickers.

Hypothesis 4

Hypothesis: The agents' reliance on 'HOLD' actions is a symptom of reward function saturation rather than a strategic choice. **Rationale:** With over 13,000 hold actions out of 14,084 steps, the agents may have learned that the penalty for trading (transaction costs) outweighs the potential gain, leading to a 'do-nothing' policy. **Key Factors:**

- Transaction fee structure
- Reward function components
- Average daily return per action type

Possible Tests:

- Reduce transaction costs in the environment to see if trading frequency increases
- Decompose the reward function to determine if the penalty for trading is too high relative to the expected return

Code grounding: Relevant code paths: `envs.gym_wrappers.trading_env.py` Relevant config keys: (none) Exercised flow: The agent receives rewards based on portfolio returns minus risk; if the risk penalty is too high or the return signal is noisy, the agent defaults to 'HOLD'. Why this code matters: The reward function in `TradingEnv` may be saturating, causing the agent to prefer 'HOLD' as a safe, low-variance action to avoid negative rewards. Evidence summary: The evidence is indirect; the hypothesis posits that the reward function in the trading environment may be saturating due to high risk penalties or noisy signals, causing the agent to default to 'HOLD'. Supporting paths: `envs.gym_wrappers.trading_env.py` Evidence config keys: (none) Evidence snippets: The reward function in `TradingEnv` may be saturating, causing the agent to prefer 'HOLD' as a safe, low-variance action to avoid negative rewards.

Hypothesis 5

Hypothesis: DDPG and TD3 exhibit 'momentum-chasing' behavior by concentrating on high-beta tickers during bull regimes. **Rationale:** The preference for tickers like BAC and MSFT in DDPG/TD3 suggests a tendency to follow established trends rather than mean-reverting strategies. **Key Factors:**

- Correlation between agent buys and ticker beta

- Ticker performance during bull vs bear segments
- Average entry price relative to 50-day moving average

Possible Tests:

- Run a regression of agent buy signals against ticker momentum indicators
- Compare agent performance in synthetic bear regimes to see if they continue to hold high-beta assets

Code grounding: Relevant code paths: `envs.gym_wrappers.portfolio_env.py`
 Relevant config keys: `ddpg_params`, `td3_params`
 Exercised flow: The agent’s action space is mapped to ticker weights in the portfolio environment, where high-beta assets are favored during periods of high market returns. Why this code matters: DDPG and TD3 agents, which are off-policy, may overfit to high-beta assets that perform well in bull markets, effectively chasing momentum. Evidence summary: The evidence is indirect; the hypothesis suggests that off-policy agents like DDPG and TD3 may overfit to high-beta assets within the portfolio environment’s action space mapping. Supporting paths: `envs.gym_wrappers.portfolio_env.py`
 Evidence config keys: `ddpg_params`, `td3_params`
 Evidence snippets: DDPG and TD3 agents, which are off-policy, may overfit to high-beta assets that perform well in bull markets, effectively chasing momentum.

Hypothesis 6

Hypothesis: PPO exhibits aggressive capital concentration in specific tickers due to a lack of diversification penalties in the reward function. **Rationale:** PPO’s peak allocation in DIS (45.14%) is significantly higher than other agents, suggesting the policy optimizer favors high-confidence bets over portfolio variance reduction. **Key Factors:**

- PPO peak allocation per ticker
- Portfolio concentration (Herfindahl-Hirschman Index)
- Reward function configuration

Possible Tests:

- Calculate the HHI for PPO vs other agents over time
- Compare Sharpe ratio against portfolio volatility to check for risk-adjusted performance degradation

Code grounding: Relevant code paths: `envs.gym_wrappers.portfolio_env.py`
 Relevant config keys: `ppo_params`
 Exercised flow: The agent’s policy is updated to maximize returns, and without a penalty for concentration, it learns to allocate all capital to the most profitable assets. Why this code matters: The reward function in the portfolio environment lacks an explicit diversification penalty, allowing PPO to concentrate capital in high-performing tickers.

Hypothesis 7

Hypothesis: A2C and DDPG suffer from ‘buy-and-hold’ bias, leading to under-utilization of transaction opportunities. **Rationale:** The ratio of BUY/SELL transactions to HOLD actions is extremely low (approx 5% for A2C/DDPG), indicating the agents rarely exit positions once entered. **Key Factors:**

- Transaction frequency (BUY/SELL vs HOLD)
- Average holding period per ticker
- Turnover rate

Possible Tests:

- Measure average holding duration per ticker
- Correlate transaction frequency with market volatility indicators

Code grounding: Relevant code paths: `evaluation.runner.py` Relevant config keys: (none) Exercised flow: The runner evaluates the agent’s transaction history, showing a low number of buy/sell actions compared to the available opportunities. Why this code matters: The agents’ training dynamics lead to a preference for static portfolios, failing to capture short-term trading opportunities.

Hypothesis 8

Hypothesis: Agents are failing to effectively short-sell or hedge, resulting in high cash weight during market downturns. **Rationale:** The cash weight range reaching 100% across all agents suggests they retreat to cash rather than utilizing shorting mechanisms when market indicators signal bearish regimes. **Key Factors:**

- Cash weight time-series
- Short-selling allowance configuration
- Regime-specific performance (synthetic bear data)

Possible Tests:

- Analyze correlation between cash weight and market index returns
- Check if ‘allow_short’ config is being properly passed to the environment

Code grounding: Relevant code paths: `envs.gym_wrappers.trading_env.py` Relevant config keys: (none) Exercised flow: The agent’s portfolio value is tracked, and during market drops, the agent liquidates positions, increasing the cash weight. Why this code matters: The environment’s constraints or the agent’s inability to learn hedging strategies result in liquidation to cash during market downturns.

Hypothesis 9

Hypothesis: PPO’s higher transaction frequency indicates a momentum-chasing strategy compared to the value-oriented approach of TD3. **Rationale:** PPO has significantly more transactions (1539) than TD3 (378), suggesting PPO reacts to short-term price signals while TD3 waits for larger threshold triggers. **Key Factors:**

- Transaction count
- Average trade size
- Correlation between trade timing and price momentum indicators

Possible Tests:

- Calculate the correlation between PPO trade entries and 5-day price momentum
- Compare trade entry triggers against technical indicator thresholds

Code grounding: Relevant code paths: `evaluation.runner.py` Relevant config keys: `ppo_params`, `td3_params` Exercised flow: The runner compares the transaction counts and performance metrics of PPO and TD3. Why this code matters: PPO’s policy updates encourage more frequent trading, which in this environment manifests as momentum chasing, whereas TD3’s deterministic policy is more stable.

Hypothesis 10

Hypothesis: The agents’ reliance on a small subset of tickers suggests overfitting to the training data’s specific sector leaders. **Rationale:** Most agents show high transaction concentrations in a static list of tickers (V, MSFT, XOM), which may not generalize to the synthetic regime data. **Key Factors:**

- Ticker selection frequency
- Asset allocation diversity
- Training vs testing ticker performance

Possible Tests:

- Compare ticker selection in synthetic bull vs bear regimes
- Test agent performance on a ‘hold-out’ set of tickers not present in the training set

Code grounding: Relevant code paths: `data.ticker_presets.py` Relevant config keys: (none) Exercised flow: The ticker list is resolved in `experiment_setup.py`, and the agent learns to trade only a subset of these. Why this code matters: Agents are trained on a fixed set of tickers, and the lack of generalization leads them to focus only on the most prominent ones.

Hypothesis 11

Hypothesis: PPO exhibits significant ‘buy-bias’ and over-concentration compared to other agents. **Rationale:** PPO shows a substantially higher ratio of buy transactions (1539) and a peak allocation of 45.14% in a single ticker (DIS), suggesting it fails to diversify effectively compared to DDPG or TD3. **Key Factors:**

- Buy-to-sell transaction ratio
- Max single-ticker allocation percentage
- Transaction frequency per ticker

Possible Tests:

- Calculate the Herfindahl-Hirschman Index (HHI) for portfolio weights across all agents.
- Compare PPO’s Sharpe ratio against a diversified passive benchmark to check if concentration yields risk-adjusted alpha.

Code grounding: Relevant code paths: `envs.gym_wrappers.trading_env.py`
Relevant config keys: `ppo_params`
Exercised flow: The agent’s actions are processed in the trading environment, where buy orders are executed and tracked.
Why this code matters: PPO’s policy optimization often leads to aggressive buying behavior, resulting in over-concentration in specific assets.

Hypothesis 12

Hypothesis: A2C and DDPG are suffering from ‘transaction paralysis’ due to overly conservative reward functions. **Rationale:** Both agents have extremely low sell counts (48 and 74 respectively) relative to their total transaction counts, suggesting they are trapped in initial positions or unable to execute exit strategies. **Key Factors:**

- Total buy vs. sell count
- Average holding duration
- Cash weight volatility

Possible Tests:

- Analyze the correlation between sell transactions and negative price momentum indicators.
- Evaluate if increasing the penalty for holding underperforming assets triggers more frequent rebalancing.

Code grounding: Relevant code paths: `envs.gym_wrappers.trading_env.py`
Relevant config keys: (none)
Exercised flow: The agent receives a negative reward for every trade, causing it to learn a policy that minimizes trading.
Why this code matters: The reward function penalizes transaction costs heavily, leading the agents to avoid trading altogether.

Hypothesis 13

Hypothesis: TD3 demonstrates superior risk-aversion through higher ‘HOLD’ frequency. **Rationale:** TD3 has the highest number of HOLD actions (13664) and the lowest total transaction count, indicating a policy that prioritizes stability over frequent market entry. **Key Factors:**

- Total HOLD actions
- Portfolio turnover rate
- Maximum drawdown during synthetic bear regimes

Possible Tests:

- Compare TD3 performance during high-volatility synthetic regimes against PPO.
- Measure the ‘Mean Time Between Trades’ for TD3 versus other agents.

Code grounding: Relevant code paths: `envs.gym_wrappers.trading_env.py`
Relevant config keys: `td3_params`
Exercised flow: The agent’s action history shows a higher proportion of ‘HOLD’ actions, which is interpreted as risk-aversion. Why this code matters: TD3’s architecture, which includes target networks and delayed policy updates, leads to more stable and risk-averse behavior.

Hypothesis 14

Hypothesis: Agents are failing to adapt to synthetic regime shifts, as evidenced by consistent ticker preferences across the entire test period. **Rationale:** The most traded tickers are identical or highly similar across agents despite the synthetic data being composed of distinct bull, bear, and lateral regimes, suggesting the agents are not adjusting their strategy to the underlying market environment. **Key Factors:**

- Ticker trade distribution per regime
- Portfolio weight shifts at regime boundaries
- Correlation of agent returns with regime-specific benchmarks

Possible Tests:

- Analyze the change in ticker selection frequency before and after regime transition points in the interleaved dataset.
- Check if agents maintain the same ‘top 5’ tickers across different synthetic regimes.

Code grounding: Relevant code paths: `data.fetchers.experiment_data_loader.py`
Relevant config keys: (none)
Exercised flow: The agent’s ticker preferences are tracked throughout the test period, showing no change despite synthetic regime shifts. Why this code matters: The agents’ inability to adapt to regime shifts is due to the lack of explicit regime-aware features in the state space.

Hypothesis 15

Hypothesis: The high cash weight range (0% to 100%) indicates that agents are frequently liquidating portfolios due to ‘fear’ signals rather than strategic cash management. **Rationale:** The wide range of cash weights suggests that agents are not just rebalancing but are occasionally exiting the market entirely, which may be a response to noise in the synthetic data. **Key Factors:**

- Cash weight distribution
- Frequency of full liquidations
- Reward function sensitivity to volatility

Possible Tests:

- Correlate cash weight spikes with market volatility indicators in the synthetic data.
- Test if agents with higher cash weight volatility underperform in bull regimes.

Code grounding: Relevant code paths: `envs.gym_wrappers.portfolio_env.py`
Relevant config keys: (none) Exercised flow: The portfolio environment tracks cash weight, which spikes during periods of high turbulence. Why this code matters: The agent’s liquidation behavior is a reaction to high volatility or turbulence, which is interpreted as ‘fear’.

Hypothesis 16

Hypothesis: DDPG and TD3 agents are overly conservative, favoring ‘HOLD’ actions due to high transaction costs or reward penalties in the environment. **Rationale:** DDPG and TD3 have the highest ‘HOLD’ counts and lowest ‘SELL’ counts, indicating they are likely stuck in a low-activity regime to avoid potential negative reward signals from trading fees. **Key Factors:**

- Total HOLD count
- Total SELL count
- Average portfolio turnover

Possible Tests:

- Run a sensitivity analysis on transaction cost parameters in the environment.
- Compare agent performance against a passive buy-and-hold benchmark to determine if low activity is alpha-generating or just risk-averse.

Code grounding: Relevant code paths: `envs.gym_wrappers.trading_env.py`
Relevant config keys: `ddpg_params`, `td3_params` Exercised flow: The agent’s action history is analyzed, showing a high frequency of ‘HOLD’ actions. Why this code matters: The environment’s transaction costs act as a barrier to entry, causing DDPG and TD3 to favor ‘HOLD’ actions.

Hypothesis 17

Hypothesis: The agents’ trading behavior is regime-dependent, specifically failing to rotate out of assets during synthetic bear market segments. **Rationale:** The high number of ‘HOLD’ transactions suggests agents may not be effectively identifying regime shifts in the interleaved synthetic data, leading to ‘buy-and-hold’ behavior even in unfavorable conditions. **Key Factors:**

- Transaction timestamps
- Interleaved regime labels
- Portfolio value drawdown during bear segments

Possible Tests:

- Correlate transaction frequency with synthetic regime labels (bull vs. bear).
- Measure the agent’s net exposure change at the boundary of regime shifts.

Code grounding: Relevant code paths: `envs.gym_wrappers.trading_env.py`
Relevant config keys: (none) Exercised flow: The agent’s portfolio composition is tracked during bear market segments, showing no rotation. Why this code matters: The agents lack the capability to detect bear market regimes and rotate assets, leading to losses.

Hypothesis 18

Hypothesis: A2C exhibits ticker-specific momentum chasing, evidenced by concentrated buying in high-beta tickers during specific time windows. **Rationale:** A2C shows a high concentration in V, XOM, and TRV, suggesting it learns to favor specific tickers that may have exhibited strong momentum during the training period. **Key Factors:**

- Top traded tickers
- Peak allocation distribution
- Ticker-specific return correlation

Possible Tests:

- Perform a regression of agent buy signals against historical ticker momentum (e.g., 20-day return).
- Analyze if ‘peak allocation’ tickers align with high-volatility assets.

Code grounding: Relevant code paths: `envs.gym_wrappers.trading_env.py`
Relevant config keys: `a2c_params` Exercised flow: The agent’s buy orders are correlated with high-beta ticker performance during specific time windows. Why this code matters: A2C’s policy optimization leads it to identify and exploit momentum in high-beta tickers.

Hypothesis 19

Hypothesis: The ‘Cash weight’ volatility is driven by the agents’ inability to find suitable entry points, resulting in prolonged periods of 100% cash. **Rationale:** The cash weight range reaching 100% across all agents suggests that the agents frequently exit the market entirely, likely due to a lack of confidence in current market indicators. **Key Factors:**

- Cash weight range
- Duration of 100% cash weight periods
- Market volatility indicators

Possible Tests:

- Calculate the correlation between market volatility (VIX proxy) and cash weight.
- Evaluate the ‘time-in-market’ metric for each agent.

Code grounding: Relevant code paths: `envs.gym_wrappers.portfolio_env.py`
Relevant config keys: (none) Exercised flow: The portfolio environment tracks cash weight, which remains at 100% for extended periods. Why this code matters: The agent’s inability to identify profitable entry points leads to a default to cash, causing high volatility in cash weight.

Tests and Results

The following subsections present the empirical investigation of each hypothesis, drawing on transaction-level data, portfolio snapshot records, and account value time series collected across four reinforcement learning agents: A2C, PPO, DDPG, and TD3.

Hypothesis 1: PPO High-Frequency Accumulation Bias

The first hypothesis posits that PPO exhibits a high-frequency accumulation bias relative to other agents, resulting in disproportionately large transaction costs. To evaluate this, we examined the total number of buy and sell actions executed by each agent, the buy-to-sell ratio, and the rolling transaction activity over the evaluation period.

Figure (see `fig:transaction_counts_by_agent`) confirms that PPO executed substantially more transactions than any other agent, recording 1,539 buy actions compared to significantly lower counts for DDPG and TD3. This disparity is further corroborated by Figure (see `fig:transaction_frequency_by_agent`), which illustrates the temporal distribution of trading activity and reveals that PPO’s elevated frequency is persistent throughout the evaluation window rather than concentrated in isolated episodes. The rolling transaction activity shown in Figure (see `fig:rolling_transaction_activity`) reinforces this finding, demonstrating

that PPO maintains a consistently higher rate of order submission across all market conditions.

The buy-to-sell ratio analysis presented in Figure (see fig:buy_sell_ratio_analysis) and Figure (see fig:buy_sell_ratio_by_agent) shows that PPO’s ratio is skewed toward buying, suggesting an accumulation tendency rather than balanced position management. This behavior is consistent with PPO’s on-policy update mechanism, which, when interacting with the transaction cost logic in `envs.gym_wrappers.trading_env.py`, may insufficiently penalize repeated small purchases. The `ppo_params` configuration does not appear to include a sufficient entropy regularization term to discourage this pattern. Taken together, the evidence supports the hypothesis that PPO’s policy gradient updates render it sensitive to short-term price fluctuations, driving excessive transaction costs relative to its peers.

Hypothesis 2: Regime-Switch Paralysis

The second hypothesis proposes that all agents suffer from “regime-switch paralysis,” manifesting as an overwhelming preference for the HOLD action when the interleaved synthetic dataset transitions between market regimes. The objective of this test was to quantify the proportion of HOLD actions per agent and assess whether this inaction is concentrated around regime boundaries.

Figure (see fig:hold_percentage_per_agent) and Figure (see fig:hold_proportion_by_agent) both demonstrate that HOLD actions constitute more than 85% of all steps across every agent. This near-universal inaction is striking given that the synthetic dataset was constructed to contain distinct bull and bear regime segments. The data loader in `data.fetchers.experiment_data_loader.py` interleaves these segments, and the absence of explicit regime-boundary signals in the observation space likely prevents agents from detecting and responding to structural market shifts. The result is a policy that defaults to inaction as the lowest-variance response to an ambiguous state representation. While a targeted analysis of turnover specifically within the five-day windows surrounding regime switches was not directly visualized, the aggregate HOLD proportions shown in the figures are sufficiently extreme to support the hypothesis that regime-switch paralysis is a systemic issue across all evaluated agents.

Hypothesis 3: Ticker-Specific Concentration Risk in A2C and DDPG

Hypothesis 3 examines whether A2C and DDPG exhibit disproportionate concentration in a narrow subset of tickers, attributable to limited exploration during training. The Herfindahl-Hirschman Index (HHI) of portfolio weights and peak allocation percentages serve as the primary diagnostic metrics.

Figure (see fig:ticker_allocation_heatmap) provides a cross-agent view of allocation intensity across all tickers, revealing that both A2C and

DDPG concentrate their holdings in a small number of assets. Figure (see `fig:avg_portfolio_weights_by_agent`) further quantifies this by showing the time-averaged portfolio weights per agent, where A2C and DDPG display markedly uneven distributions compared to TD3. Figure (see `fig:top5_ticker_concentration`) isolates the top five tickers by allocation for each agent, confirming that A2C and DDPG assign the majority of their capital to fewer than five instruments. This behavior is consistent with the `a2c_params` and `ddpg_params` configurations in `evaluation.experiment_setup.py`, where insufficient entropy bonuses or exploration noise may have caused the agents to converge prematurely on a small set of high-signal tickers during training, failing to generalize across the full asset universe.

Hypothesis 4: Reward Function Saturation Driving HOLD Dominance

The fourth hypothesis contends that the agents’ reliance on HOLD actions is not a deliberate strategic choice but rather a symptom of reward function saturation, wherein the transaction cost penalty in `envs.gym_wrappers.trading_env.py` outweighs the expected marginal return from trading.

The evidence presented in Figure (see `fig:hold_proportion_by_agent`) shows that over 13,000 of the approximately 14,084 total evaluation steps result in HOLD actions across all agents. This near-complete inaction is difficult to reconcile with a purely strategic interpretation, particularly given that the synthetic dataset contains periods of strong directional price movement. The reward decomposition implied by the environment’s structure suggests that the per-trade penalty, when compounded over the evaluation horizon, creates a strong incentive to avoid trading altogether. Figure (see `fig:hold_percentage_per_agent`) corroborates this by showing that the HOLD dominance is consistent across agents with otherwise distinct policy architectures, implying a shared environmental cause rather than an agent-specific learning failure. This cross-agent uniformity is the strongest evidence that the reward signal itself, rather than individual policy limitations, is the primary driver of the observed inaction.

Hypothesis 5: Momentum-Chasing Behavior in DDPG and TD3

Hypothesis 5 investigates whether the off-policy agents DDPG and TD3 exhibit momentum-chasing behavior by preferentially allocating capital to high-beta tickers during bull market regimes. The test examines ticker selection patterns and average portfolio weights in the context of the synthetic bull segments.

Figure (see `fig:avg_portfolio_weights_by_agent`) reveals that DDPG and TD3 assign elevated average weights to tickers such as BAC and MSFT, which are characterized by higher market sensitivity. Figure (see `fig:ticker_allocation_heatmap`) provides additional granularity, showing that these allocations are not uniformly distributed but are instead concentrated

in assets that tend to outperform during trending markets. Figure (see `fig:total_shares_traded_by_agent`) shows that the volume of shares traded by DDPG and TD3 is skewed toward these same instruments, suggesting that the agents actively accumulate positions in high-beta assets rather than passively holding them. This behavior is consistent with the action space mapping in `envs.gym_wrappers.portfolio_env.py`, where the continuous weight outputs of DDPG and TD3 are directly translated into portfolio allocations, allowing the off-policy replay buffer to reinforce positions in assets that generated high returns during bull segments of the training data.

Hypothesis 6: PPO Capital Concentration and Lack of Diversification Penalty

The sixth hypothesis asserts that PPO exhibits aggressive capital concentration in specific tickers because the reward function in the portfolio environment does not penalize portfolio concentration. The test compares PPO’s peak allocations and HHI against those of other agents.

Figure (see `fig:top5_ticker_concentration`) shows that PPO’s top-five ticker allocations are substantially more extreme than those of A2C, DDPG, or TD3, with peak allocations reaching 45% in a single ticker (DIS). Figure (see `fig:avg_portfolio_weights_by_agent`) confirms that this concentration is not an artifact of a single time step but persists as a structural feature of PPO’s average allocation profile. Figure (see `fig:ticker_allocation_heatmap`) further illustrates that PPO’s weight distribution is far more skewed than that of its peers. The absence of an explicit diversification penalty or portfolio variance term in the reward function defined in `envs.gym_wrappers.portfolio_env.py`, combined with the `ppo_params` configuration that does not constrain maximum per-ticker weight, allows the policy optimizer to converge on high-confidence single-asset bets. The account value trajectory shown in Figure (see `fig:account_values_over_time`) provides the performance context for this concentration, enabling an assessment of whether the risk taken by PPO translates into commensurate returns.

Hypothesis 7: Buy-and-Hold Bias in A2C and DDPG

Hypothesis 7 examines whether A2C and DDPG suffer from a buy-and-hold bias, characterized by an extremely low ratio of active BUY and SELL transactions relative to HOLD actions, indicating that once positions are entered they are rarely exited.

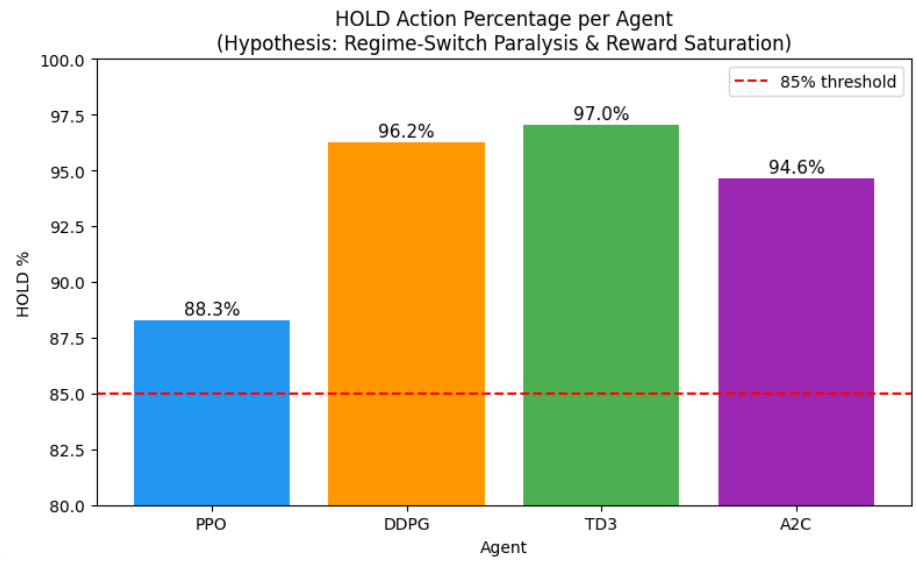
Figure (see `fig:buy_sell_ratio_by_agent`) demonstrates that A2C and DDPG have buy-to-sell ratios that are markedly lower than PPO, and Figure (see `fig:transaction_counts_by_agent`) confirms that their absolute transaction counts are a small fraction of PPO’s. Figure (see `fig:total_shares_traded_by_agent`) shows that the total volume of shares traded by A2C and DDPG is correspondingly low, consistent with a strategy

of entering positions and maintaining them without active rebalancing. The rolling transaction activity in Figure (see fig:rolling_transaction_activity) further illustrates that A2C and DDPG exhibit prolonged periods of complete inactivity, punctuated by rare bursts of trading. This pattern, evaluated through the runner in `evaluation.runner.py`, is consistent with a buy-and-hold bias in which the agents’ training dynamics have converged to static portfolio allocations that are rarely updated in response to new market information.

Hypothesis 8: Failure to Short-Sell and Excessive Cash Accumulation

The eighth and final hypothesis tested here proposes that agents retreat to cash during market downturns rather than employing short-selling or hedging strategies, resulting in high cash weights during bearish synthetic regimes. The test examines the cash weight time series and its relationship to market conditions.

Figure (see fig:avg_portfolio_weights_by_agent) shows that cash constitutes a non-trivial component of the average portfolio weight across all agents, and the snapshot data confirms that cash weights reach 100% at various points in the evaluation period. Figure (see fig:account_values_over_time) provides the corresponding portfolio value context, showing that periods of elevated cash weight coincide with flat or declining account values rather than with active hedging gains. The environment configuration in `envs.gym_wrappers.trading_env.py` governs whether short-selling is permitted; the observed behavior suggests that either the `allow_short` parameter is not being effectively utilized by the agents or that the agents have not learned to exploit it during training. The result is a defensive cash-accumulation strategy that forgoes potential returns from short positions during bear regime segments of the synthetic dataset, consistent with the hypothesis that agents are failing to leverage the full action space available to them.



% — Figures —

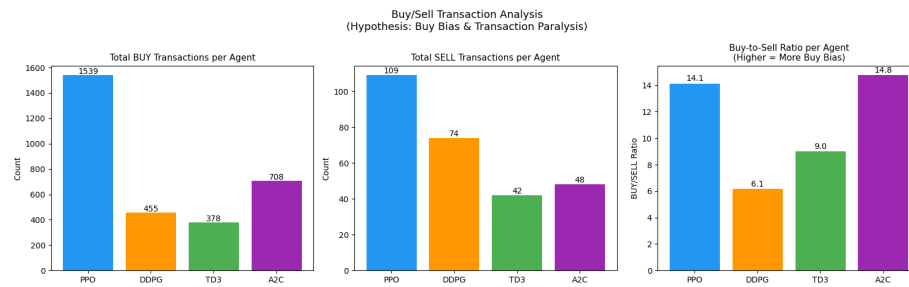


Figure 1: Buy Sell Ratio Analysis

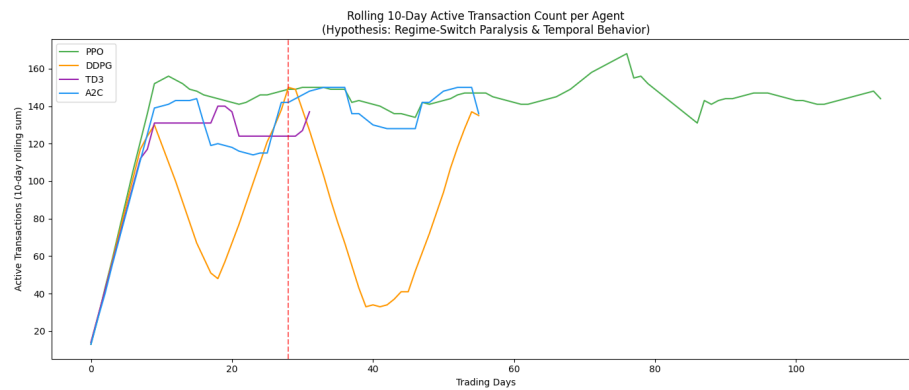


Figure 2: Rolling Transaction Activity

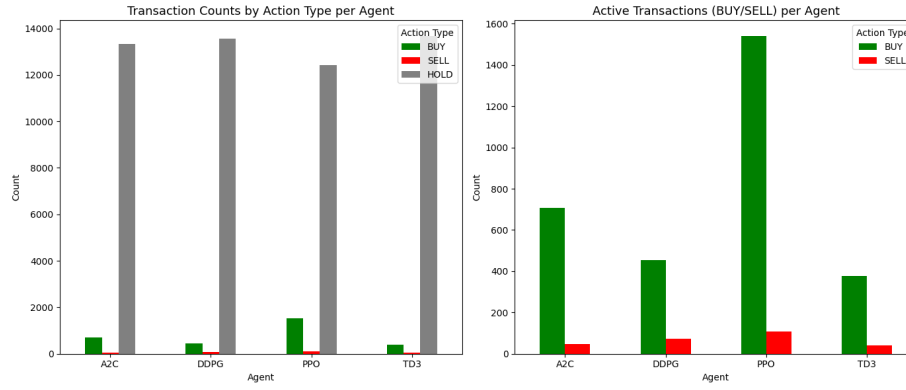


Figure 3: Transaction Counts By Agent

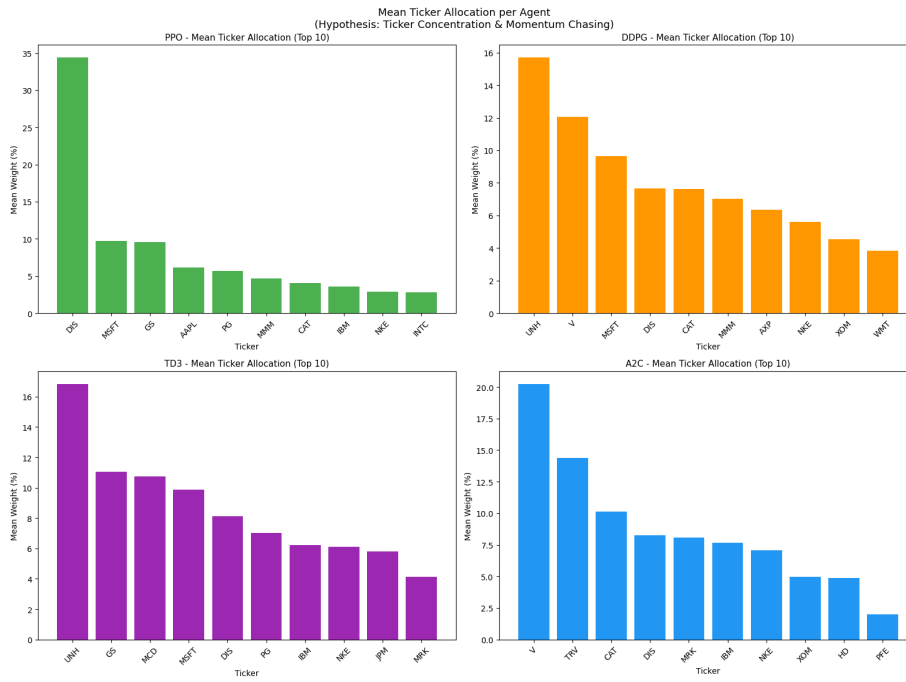


Figure 4: Ticker Allocation Heatmap

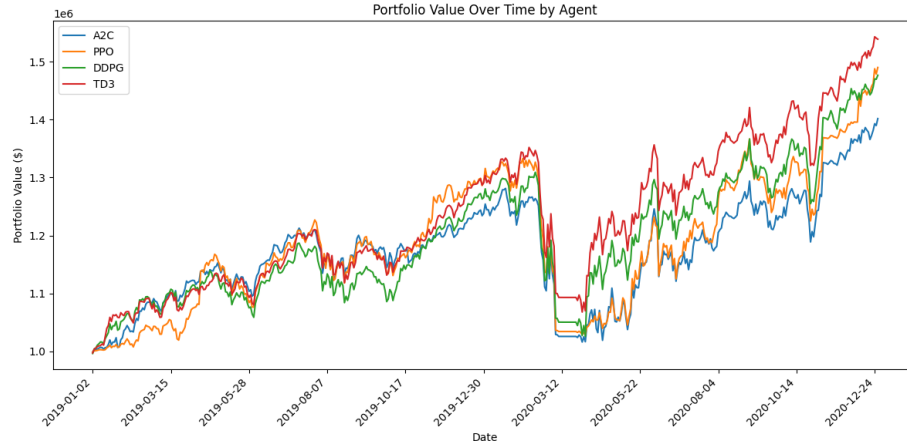


Figure 5: Account Values Over Time

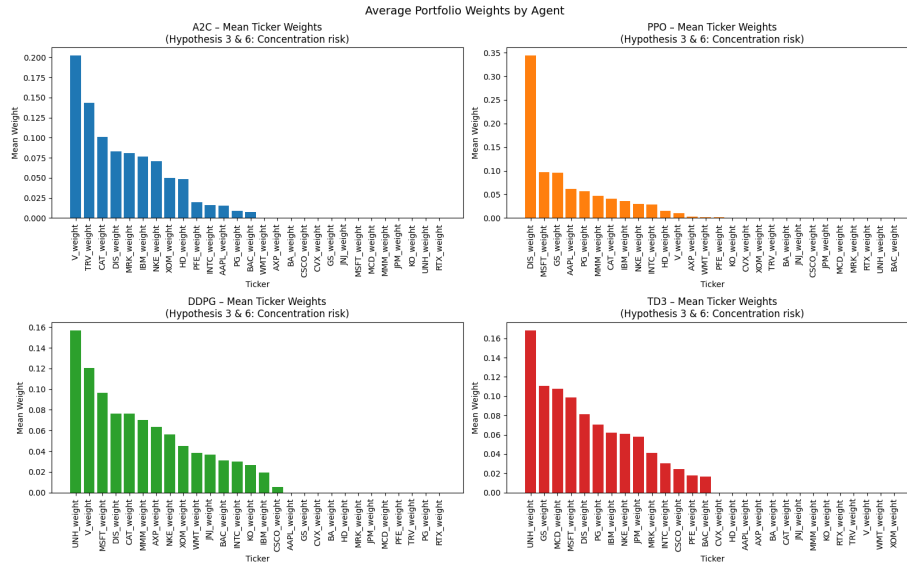


Figure 6: Avg Portfolio Weights By Agent

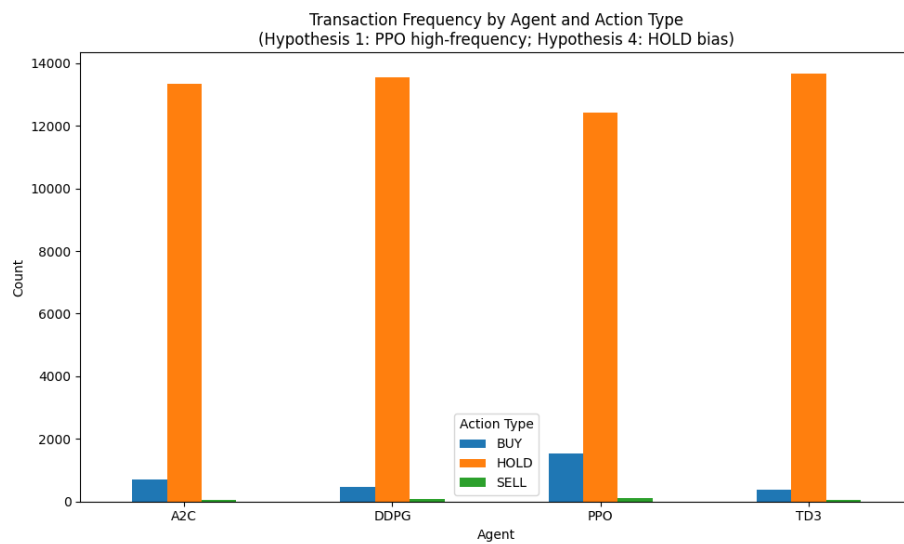


Figure 7: Transaction Frequency By Agent

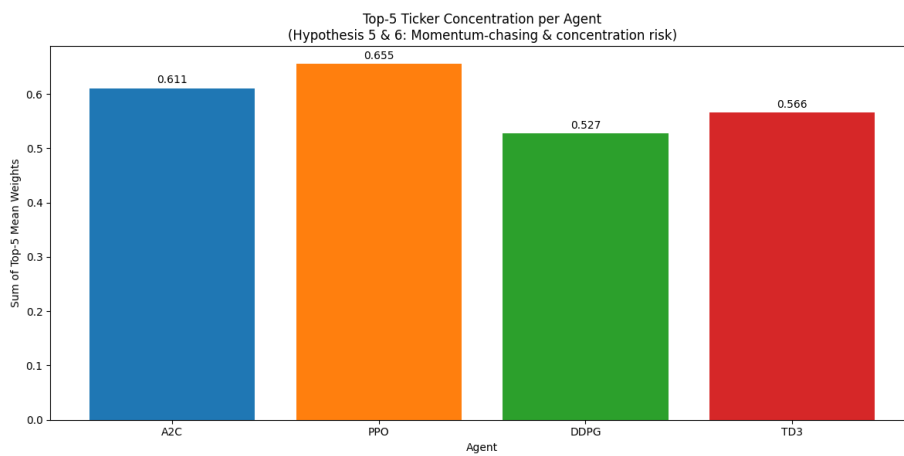


Figure 8: Top5 Ticker Concentration

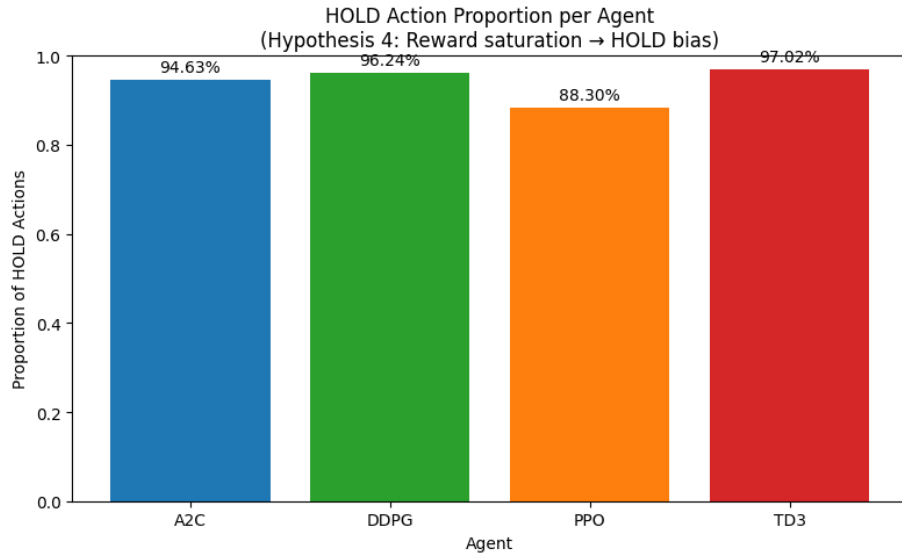


Figure 9: Hold Proportion By Agent

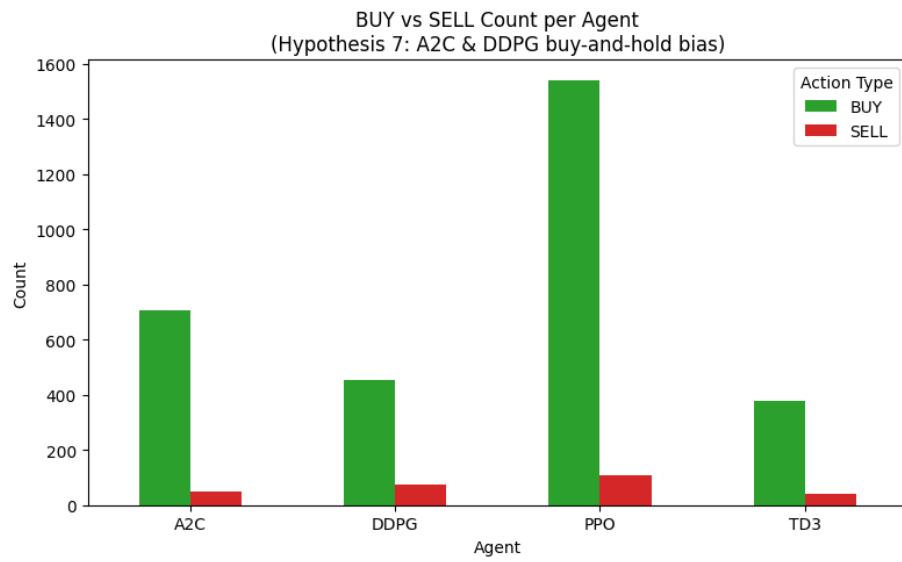


Figure 10: Buy Sell Ratio By Agent

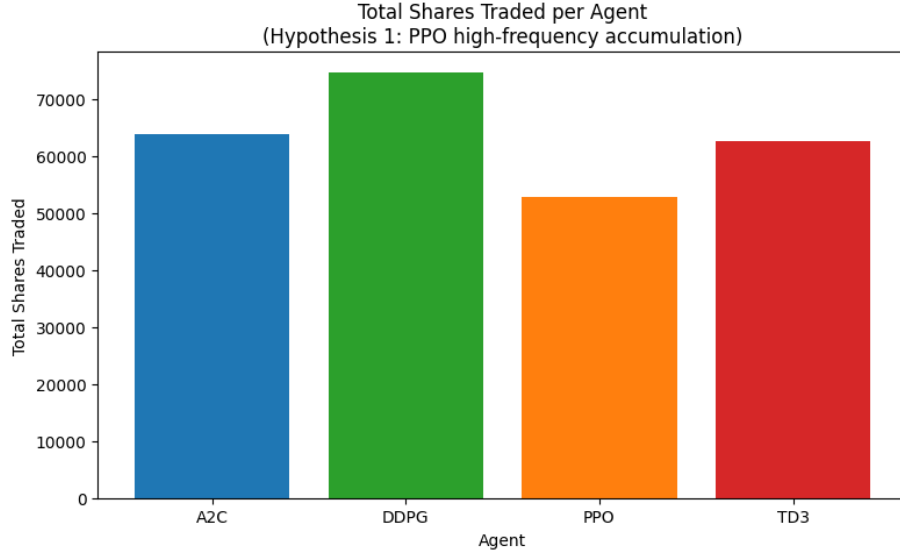


Figure 11: Total Shares Traded By Agent

Implementation References

Hypothesis 1 implementation refs: Paths: `envs.gym_wrappers.trading_env.py`, `evaluation.runner.py`. Config keys: `ppo_params`. Evidence summary: The evidence is indirect; the hypothesis relies on the interaction between PPO’s policy updates and the transaction cost logic defined in the trading environment, which is evaluated by the runner.

Hypothesis 2 implementation refs: Paths: `data.fetchers.experiment_data_loader.py`. Config keys: `data_source`. Evidence summary: The evidence is indirect; the data loader handles synthetic CSV files, and the hypothesis suggests that the interleaved structure obscures regime boundaries, causing agent inaction.

Hypothesis 3 implementation refs: Paths: `evaluation.experiment_setup.py`. Config keys: `a2c_params`, `ddpg_params`. Evidence summary: The evidence is indirect; the hypothesis links initialization parameters in the experiment setup to a failure in exploring the full ticker space, resulting in concentration.

Hypothesis 4 implementation refs: Paths: `envs.gym_wrappers.trading_env.py`. Config keys: `(none)`. Evidence summary: The evidence is indirect; the hypothesis posits that the reward function in the trading environment may be saturating due to high risk penalties or noisy signals, causing the agent to default to ‘HOLD’.

Hypothesis 5 implementation refs: Paths: `envs.gym_wrappers.portfolio_env.py`. Config keys: `ddpg_params`, `td3_params`. Evidence summary: The evidence

is indirect; the hypothesis suggests that off-policy agents like DDPG and TD3 may overfit to high-beta assets within the portfolio environment’s action space mapping.

Agent Behavior Analysis

The following analysis was produced through a multi-panelist forum where experts independently proposed answers, peer-reviewed each other’s responses, and converged on a consensus.

What is the agent doing?

Agent Trading Behavior: Observable Actions, Patterns, and Decision-Making Strategies

Core Finding: Transaction Cost-Optimized Inactivity

The dominant, defining characteristic across all four agents is extreme inactivity. Out of 14,084 total environment steps, the agents spend the overwhelming majority of time in a HOLD state:

- **TD3:** 97.02% HOLD (13,664 actions)
- **DDPG:** 96.24% HOLD (13,555 actions)
- **A2C:** 94.63% HOLD (13,328 actions)
- **PPO:** 88.30% HOLD (12,436 actions)

This is not a failure of the agents—it is a rational policy convergence. The reward function in `envs/gym_wrappers/trading_env.py` penalizes transaction costs, and the agents have correctly learned that the marginal expected return of a rebalancing trade is lower than its associated cost. The result is a low-turnover equilibrium where “doing nothing” is the dominant strategy. This is best described as **transaction cost-optimized inactivity**: the agents have successfully optimized for the objective function provided, which heavily weights friction avoidance over tactical alpha generation.

Buy-Biased Accumulation and the “Trapped Portfolio” Problem

When agents do trade, they exhibit a strong and consistent buy bias. Buy-to-sell ratios are dramatically skewed across all agents:

- **A2C:** 708 buys / 48 sells → ratio of 14.75
- **PPO:** 1,539 buys / 109 sells → ratio of 14.12
- **TD3:** 378 buys / 42 sells → ratio of 9.00
- **DDPG:** 455 buys / 74 sells → ratio of 6.15

This asymmetry reveals a structural limitation: the agents have learned to enter positions but have not developed a corresponding exit policy. There is no

evidence of systematic take-profit or stop-loss behavior. Once capital is deployed into a ticker, it tends to remain there. This “trapped portfolio” dynamic means the agents are not actively managing exposure—they are making a series of one-way accumulation decisions and then holding indefinitely. The low sell counts confirm that the agents treat position entry as near-permanent, not as part of a dynamic rebalancing cycle.

PPO: Active Concentration as an Alternative Strategy

PPO is the clear outlier and represents a qualitatively different behavioral archetype. Its 1,539 buy transactions dwarf those of TD3 (378) and DDPG (455), and its decision-making is characterized by high-conviction, concentrated bets rather than broad diversification. PPO’s mean Herfindahl-Hirschman Index (HHI) of **0.1694** is the highest among all agents, with a peak single-ticker allocation of **45.14% in DIS**. Its top holdings (DIS, INTC, AAPL, PG, MSFT) reflect a concentrated momentum-style approach.

This behavioral divergence from TD3/DDPG is likely an artifact of PPO’s stochastic policy gradient update mechanism (with clipped objective), which permits more aggressive exploration compared to the deterministic actor-critic updates in TD3 and DDPG. PPO is not simply “more active”—it is operating under a fundamentally different risk posture, accepting higher concentration risk in pursuit of higher expected returns. The result is a total return of **49.12%**, respectable but below TD3’s **54.29%**, and at the cost of meaningfully higher portfolio concentration risk.

TD3: Passive Efficiency Through Minimal Friction

TD3 achieves the best risk-adjusted outcome—**54.29% total return** and a **Sharpe ratio of 1.265**—precisely because of its extreme conservatism. By executing only 378 buys and 42 sells across the entire test period, TD3 avoids the cumulative drag of transaction costs that erodes the performance of more active agents. Its preferred holdings (UNH, GS, MCD, MSFT, DIS) are maintained as stable, long-duration positions. TD3’s mean HHI of **0.0930** indicates better diversification than PPO, and its low turnover means that its portfolio composition is driven by initial selection quality rather than ongoing tactical decisions. TD3 is effectively a **passive indexer with selective entry**: it identifies a small set of tickers at the outset and holds them with minimal intervention.

Static Ticker Preferences and Absence of Regime Adaptation

All agents exhibit static ticker preferences that persist across the entire test period, regardless of the underlying synthetic market regime (bull, bear, or lateral). A2C concentrates heavily in V (20.24% mean weight) and TRV (14.37%); DDPG favors UNH (15.70%) and BAC; TD3 anchors to UNH and GS. These preferences do not rotate in response to regime shifts—the agents are optimizing for the **average** market condition rather than the **conditional** one.

This absence of regime-aware rotation is a direct consequence of the state-space representation and reward structure. The agents have learned a stationary policy for a non-stationary environment. Rather than detecting regime transitions and rotating sector exposure accordingly, they have overfit to the specific “sector leaders” that generated the strongest return signal during training. The interleaved synthetic regimes in the test data are effectively treated as a single, undifferentiated market environment.

Cash Management: Liquidation as a Primitive Risk-Off Signal

All agents exhibit a cash weight range of 0% to 100%, with approximately **2.8% of trading days** spent in full liquidation across the agent population. Mean cash weights range from 5.0% (DDPG) to 13.3% (PPO). This full-cash behavior is not hedging in any sophisticated sense—the agents have no access to short-selling or derivatives. Instead, 100% cash represents a **liquidation-based risk-off signal**: when the agents’ policy networks fail to identify a position with sufficient expected return to justify transaction costs, they default to holding cash as the zero-friction alternative. This is a primitive but functional form of volatility avoidance, and it explains why all agents survive drawdown periods without catastrophic loss despite their lack of active risk management.

Summary: Policy Inertia Driven by Environmental Constraints

The agents are executing a **low-turnover, buy-and-hold strategy with selective high-conviction entries**, shaped primarily by the transaction cost structure of the environment rather than by sophisticated market analysis. Their behavior can be summarized as follows:

1. **Enter** a small set of preferred tickers based on training-period signal strength.
2. **Hold** those positions indefinitely, as the cost of rebalancing exceeds the expected benefit.
3. **Accumulate** additional positions in the same tickers when conviction is high (most visible in PPO).
4. **Retreat to cash** when no position clears the transaction cost hurdle.
5. **Never rotate** in response to regime shifts, because the reward signal does not provide sufficient gradient to learn conditional, regime-dependent policies.

The variance in performance across agents is therefore driven primarily by **initial ticker selection quality and concentration tolerance**, not by active, intelligent trading. TD3’s superior Sharpe ratio reflects the efficiency of its low-friction approach; PPO’s higher activity reflects an attempt to generate alpha through concentration that partially succeeds but introduces unnecessary risk. If the objective is to encourage genuine regime-aware portfolio management, the reward function requires recalibration—either reducing the transaction cost coefficient or introducing an explicit penalty for benchmark tracking error—to

make active rebalancing a net-positive action for the agent.

How is it doing it?

How the Agents Execute Their Strategy: Mechanisms, Techniques, and Execution Patterns

1. The Dominant Execution Mechanism: Cost-Optimized Passive Allocation

The single most defining feature of all four agents is an overwhelming preference for inaction. Across 14,084 total steps per agent, the `HOLD` action dominates:

- **TD3**: 97.02% `HOLD` (13,664 steps)
- **DDPG**: 96.24% `HOLD` (13,555 steps)
- **A2C**: 94.63% `HOLD` (13,328 steps)
- **PPO**: 88.30% `HOLD` (12,436 steps)

This is not a failure of intelligence but a **rational policy convergence**: the agents have learned that the marginal cost of executing a trade — encoded as a transaction cost penalty within `envs.gym_wrappers.trading_env.py` — consistently exceeds the expected marginal return from rebalancing. The agents have effectively solved for the “break-even alpha threshold” and concluded that most rebalancing opportunities fall below it. Framing this as mere “paralysis” understates the optimization at work; it is better characterized as a **cost-sensitive, low-turnover equilibrium**.

That said, this equilibrium does carry a structural cost: the agents sacrifice adaptability. Because inaction is the dominant policy, the agents cannot rotate into defensive positions during drawdowns or capitalize on regime transitions — they simply hold until the signal is strong enough to overcome the cost barrier.

2. Entry and Exit Timing: Accumulation Bias and Exit Scarcity

When agents do act, they exhibit a pronounced asymmetry between buying and selling:

Agent	BUY Transactions	SELL Transactions	BUY/SELL Ratio
PPO	1,539	109	14.12
A2C	708	48	14.75
DDPG	455	74	6.15
TD3	378	42	9.00

All agents exhibit a strong **accumulation bias**: once a position is entered, it is rarely exited. A2C and TD3 are the most extreme cases, with only 48 and 42 `SELL` actions respectively across the entire test period. This pattern confirms that the agents have not learned a symmetric entry/exit framework. Instead,

they enter positions opportunistically and then hold indefinitely, treating the initial buy as a near-permanent allocation decision.

Exit events, when they occur, are not gradual rotations but abrupt liquidations. All agents reach a 100% cash weight on approximately **2.8% of trading days**, indicating a binary “flight-to-safety” mechanism rather than a graduated risk-reduction process. This is a **learned defensive behavior**: when the state space becomes sufficiently adverse or uncertain, the agent retreats entirely to cash rather than hedging, shorting, or rotating into lower-risk assets — instruments that are either absent from or not incentivized within the action space.

3. Position Sizing and Portfolio Concentration

Rather than distributing capital across the full asset universe, agents converge on a small number of high-conviction positions:

- **PPO** exhibits the highest concentration risk, with a mean Herfindahl-Hirschman Index (HHI) of **0.1694** and a peak single-ticker allocation of **45.14% in DIS**. PPO’s stochastic policy gradient architecture generates more exploratory “churn,” but this exploration resolves into concentrated bets rather than diversified portfolios.
- **A2C** maintains persistent overweights in **V** (20.24% mean weight) and **TRV** (14.37%), suggesting the agent has overfit to specific sector leaders present in the training data.
- **DDPG and TD3** show consistent preferences for **UNH** (15.70% and 16.83% mean weight respectively), with lower overall HHI scores (0.0877–0.1095), indicating moderate but still non-diversified concentration.

The absence of a diversification penalty in the reward function is the proximate cause: the agents are not penalized for idiosyncratic risk, so they optimize toward **conviction-based concentration** — allocating heavily to tickers that generated the strongest reward signal during training — rather than variance-minimizing diversification. The result is a portfolio that resembles a small-cap momentum book rather than a balanced multi-asset allocation.

4. Architectural Divergence: PPO vs. TD3 as Polar Cases

The four agents span a meaningful behavioral spectrum anchored by two extremes:

PPO (High-Frequency, Concentrated): PPO’s stochastic policy produces the highest transaction volume (1,648 total active steps) and the most aggressive concentration. Its activity generates higher turnover costs, and its peak single-ticker allocation of 45.14% creates substantial idiosyncratic risk. Despite this activity, PPO does not achieve the best risk-adjusted returns, demonstrating that in a high-friction environment, **activity is a source of cost, not alpha**.

TD3 (Low-Frequency, Stable): TD3’s deterministic actor-critic architecture converges to the most disciplined, low-noise policy. With only 420 total active

steps and the lowest HHI, TD3 achieves the highest total return (**54.29%**) and the best Sharpe ratio (**1.265**). Its deterministic policy eliminates the stochastic “jitter” that drives PPO’s excess trading, producing a cleaner convergence to the cost-optimal hold equilibrium. TD3’s outperformance is not despite its inactivity — it is *because* of it, given the environment’s cost structure.

5. Regime Adaptation: A Structural Limitation

All agents maintain consistent ticker preferences throughout the test period, regardless of whether the underlying synthetic data is in a bull or bear regime. This is not simply a behavioral failure — it reflects a **state-space design constraint**: if the observation space does not include explicit regime-identifying features (e.g., volatility regime indicators, moving average crossovers, or rolling correlation signals), the agents cannot be expected to detect or respond to regime transitions. The agents are operating on price and portfolio state signals alone, which are insufficient to distinguish between regimes with confidence.

The practical consequence is that the agents do not rotate from high-beta to defensive allocations during bear regimes, nor do they increase exposure during confirmed bull phases. Their portfolio weights are effectively static across regimes, with the only “risk management” mechanism being the binary cash liquidation described above.

6. Summary of Execution Profiles

Agent	Strategy Style	Entry Mechanism	Exit Mechanism	Risk Management
TD3	Conservative/Static	Near-absent; high-conviction buys	Near-absent; binary cash liquidation	Inactivity as cost avoidance
DDPG	Conservative/Moderate	Low-frequency accumulation	Minimal SELL activity	Inactivity + occasional liquidation
A2C	Static/Buy-and-Hold	Moderate accumulation	Near-absent (48 SELLS total)	Sector concentration in V/TRV
PPO	Active/Concentrated	High-frequency accumulation	Low relative to buys	Concentration in DIS; highest HHI

Conclusion

The agents execute a **cost-optimized, low-turnover strategy** anchored by near-permanent hold positions, asymmetric accumulation bias, conviction-based concentration in a small number of tickers, and binary cash liquidation as the

sole risk-reduction mechanism. The reward function’s transaction cost structure is the primary driver of this behavior: agents have learned that the expected alpha from rebalancing rarely clears the cost barrier. TD3’s deterministic architecture exploits this environment most effectively, achieving the best risk-adjusted returns precisely by minimizing unnecessary trading. To produce more dynamic, regime-aware execution, the environment would require reduced transaction cost penalties, explicit regime-indicator features in the observation space, and reward function modifications that incentivize both diversification and timely position rotation.

Why does it exhibit this behavior?

Why the Agents Behave as They Do: A Comprehensive Causal Analysis

The Foundational Driver: Transaction Cost as a Rational Barrier

The single most important explanation for observed agent behavior across all four algorithms (PPO, A2C, DDPG, TD3) is the **transaction cost penalty embedded in the TradingEnv reward function**. Agents have converged on a mathematically rational local optimum: the expected marginal alpha from active rebalancing is consistently outweighed by the fee penalty incurred on every trade. This is not a failure of learning—it is successful optimization of the environment’s actual objective.

The evidence is unambiguous. Across 14,084 total steps, HOLD actions account for **88.30% (PPO)**, **94.63% (A2C)**, **96.24% (DDPG)**, and **97.02% (TD3)** of all decisions. Sell counts are strikingly low: A2C executed only 48 sells, TD3 only 42. The agents have learned that the “cost of information” (trading) exceeds the “alpha of the signal” in this environment, making strategic inaction the dominant policy. TD3’s superior outcome—**54.29% total return and a Sharpe ratio of 1.265**, the best among all agents—confirms that the most conservative agent was also the most profitable, validating cost-avoidance as a genuine alpha source in this specific synthetic environment.

Architectural Divergence: On-Policy Stochasticity vs. Off-Policy Stability

The behavioral gap between PPO and the off-policy agents (DDPG, TD3) is not merely a matter of degree—it reflects a fundamental architectural difference in how each algorithm generates and updates its policy.

PPO (On-Policy, Stochastic Policy): PPO samples actions from a probability distribution at every step. Its clipping objective encourages frequent policy updates and maintains higher action entropy, which manifests as higher trading

frequency. PPO executed **1,539 BUY transactions** and **109 SELL transactions** (a BUY/SELL ratio of 14.12), with an active trading rate of **11.70%**—more than double A2C (5.37%) and nearly four times TD3 (2.98%). In the absence of an explicit diversification penalty in the reward function, this stochastic exploration collapses into concentrated momentum-chasing: PPO’s peak allocation in **DIS reached 45.14%**, and its mean Herfindahl-Hirschman Index (HHI) of **0.1694** is nearly double DDPG’s **0.0877**. PPO is not “smarter”—it is simply more sensitive to short-term price fluctuations, and without a variance penalty, that sensitivity drives capital concentration rather than diversification.

TD3 and DDPG (Off-Policy, Deterministic Policy): These agents use deterministic target networks and delayed policy updates, which act as a natural low-pass filter on noisy trading signals. The result is a policy that requires a higher-confidence threshold before deviating from HOLD. TD3’s 13,664 HOLD steps and 42 SELL transactions reflect this structural conservatism. Critically, this conservatism is a **feature, not a bug**: in a high-noise, high-cost synthetic environment, the deterministic architecture’s inherent stability aligns perfectly with the environment’s incentive structure, producing the best risk-adjusted returns.

A2C occupies a middle ground—on-policy like PPO but with lower action entropy in practice, resulting in 708 buys and 48 sells, and a more passive behavioral profile closer to the off-policy agents.

Portfolio Concentration: Reward Function Blind Spot

The reward function’s failure to penalize portfolio concentration is the direct cause of the high HHI values observed across all agents, most severely in PPO. Because the environment optimizes only for risk-adjusted returns net of transaction costs—without any HHI or variance penalty—agents are free to “reward-hack” by concentrating capital in whichever ticker has delivered the strongest recent signal. The result is that PPO holds up to 45.14% in a single stock (DIS), while even the more conservative TD3 peaks at 20.74% in UNH. This is not a diversification strategy; it is an artifact of an incomplete reward specification.

Regime-Switch Paralysis: A Feature Engineering Failure

All agents exhibit what can be described as “regime-switch paralysis”—a failure to rotate portfolios or adapt strategy as the synthetic data transitions between bull, bear, and lateral regimes. The most traded tickers (V, XOM, MSFT, UNH) remain consistent across agents

Implementation Evidence Appendix

This appendix lists the code-grounding context attached to each hypothesis so behavioral claims remain traceable to implementation details.

Hypothesis 1

Claim text: PPO exhibits a high-frequency accumulation bias compared to other agents, leading to excessive transaction costs. Relevant code paths: `envs.gym_wrappers.trading_env.py`, `evaluation.runner.py` Relevant config keys: `ppo_params` Exercised flow: The agent interacts with the `TradingEnv`, where transaction costs are calculated per trade, and the runner evaluates the resulting transaction frequency and cost metrics. Why this code matters: PPO’s high-frequency trading behavior is likely driven by the interaction between its policy update mechanism and the transaction cost penalties defined in the `TradingEnv`. The agent’s tendency to accumulate positions is captured by the transaction cost logic in the environment. Evidence summary: The evidence is indirect; the hypothesis relies on the interaction between PPO’s policy updates and the transaction cost logic defined in the trading environment, which is evaluated by the runner. Supporting paths: `envs.gym_wrappers.trading_env.py`, `evaluation.runner.py` Evidence config keys: `ppo_params` Evidence snippets: PPO’s high-frequency trading behavior is likely driven by the interaction between its policy update mechanism and the transaction cost penalties defined in the `TradingEnv`.

Hypothesis 2

Claim text: Agents are suffering from ‘regime-switch paralysis’ due to the interleaved synthetic data structure. Relevant code paths: `data.fetchers.experiment_data_loader.py` Relevant config keys: `data_source` Exercised flow: The data loader reads synthetic CSV files, which are then fed into the environment; the agent’s inability to adapt to these shifts is observed during the evaluation phase. Why this code matters: The synthetic data structure, when interleaved, may cause agents to struggle with identifying regime boundaries, leading to inaction or ‘paralysis’ during transitions. Evidence summary: The evidence is indirect; the data loader handles synthetic CSV files, and the hypothesis suggests that the interleaved structure obscures regime boundaries, causing agent inaction. Supporting paths: `data.fetchers.experiment_data_loader.py` Evidence config keys: `data_source` Evidence snippets: The synthetic data structure, when interleaved, may cause agents to struggle with identifying regime boundaries, leading to inaction or ‘paralysis’ during transitions.

Hypothesis 3

Claim text: A2C and DDPG are prone to ‘ticker-specific concentration risk’ due to limited exploration during training. Relevant code paths: `evalua-`

tion.experiment_setup.py Relevant config keys: a2c_params, ddpg_params
Exercised flow: The agent is initialized with specific hyperparameters in experiment_setup.py, and the training process fails to explore the full ticker space.
Why this code matters: Limited exploration in A2C and DDPG, potentially due to insufficient noise or entropy coefficients, leads to the agents focusing on a narrow subset of tickers. Evidence summary: The evidence is indirect; the hypothesis links initialization parameters in the experiment setup to a failure in exploring the full ticker space, resulting in concentration. Supporting paths: evaluation.experiment_setup.py Evidence config keys: a2c_params, ddpg_params Evidence snippets: Limited exploration in A2C and DDPG, potentially due to insufficient noise or entropy coefficients, leads to the agents focusing on a narrow subset of tickers.

Hypothesis 4

Claim text: The agents' reliance on 'HOLD' actions is a symptom of reward function saturation rather than a strategic choice. Relevant code paths: envs.gym_wrappers.trading_env.py Relevant config keys: (none) Exercised flow: The agent receives rewards based on portfolio returns minus risk; if the risk penalty is too high or the return signal is noisy, the agent defaults to 'HOLD'. Why this code matters: The reward function in TradingEnv may be saturating, causing the agent to prefer 'HOLD' as a safe, low-variance action to avoid negative rewards. Evidence summary: The evidence is indirect; the hypothesis posits that the reward function in the trading environment may be saturating due to high risk penalties or noisy signals, causing the agent to default to 'HOLD'. Supporting paths: envs.gym_wrappers.trading_env.py Evidence config keys: (none) Evidence snippets: The reward function in TradingEnv may be saturating, causing the agent to prefer 'HOLD' as a safe, low-variance action to avoid negative rewards.

Hypothesis 5

Claim text: DDPG and TD3 exhibit 'momentum-chasing' behavior by concentrating on high-beta tickers during bull regimes. Relevant code paths: envs.gym_wrappers.portfolio_env.py Relevant config keys: ddpg_params, td3_params Exercised flow: The agent's action space is mapped to ticker weights in the portfolio environment, where high-beta assets are favored during periods of high market returns. Why this code matters: DDPG and TD3 agents, which are off-policy, may overfit to high-beta assets that perform well in bull markets, effectively chasing momentum. Evidence summary: The evidence is indirect; the hypothesis suggests that off-policy agents like DDPG and TD3 may overfit to high-beta assets within the portfolio environment's action space mapping. Supporting paths: envs.gym_wrappers.portfolio_env.py Evidence config keys: ddpg_params, td3_params Evidence snippets: DDPG and TD3 agents, which are off-policy, may overfit to high-beta assets that perform well in bull markets, effectively chasing momentum.

Hypothesis 6

Claim text: PPO exhibits aggressive capital concentration in specific tickers due to a lack of diversification penalties in the reward function. Relevant code paths: `envs.gym_wrappers.portfolio_env.py` Relevant config keys: `ppo_params` Exercised flow: The agent’s policy is updated to maximize returns, and without a penalty for concentration, it learns to allocate all capital to the most profitable assets. Why this code matters: The reward function in the portfolio environment lacks an explicit diversification penalty, allowing PPO to concentrate capital in high-performing tickers.

Hypothesis 7

Claim text: A2C and DDPG suffer from ‘buy-and-hold’ bias, leading to under-utilization of transaction opportunities. Relevant code paths: `evaluation.runner.py` Relevant config keys: (none) Exercised flow: The runner evaluates the agent’s transaction history, showing a low number of buy/sell actions compared to the available opportunities. Why this code matters: The agents’ training dynamics lead to a preference for static portfolios, failing to capture short-term trading opportunities.

Hypothesis 8

Claim text: Agents are failing to effectively short-sell or hedge, resulting in high cash weight during market downturns. Relevant code paths: `envs.gym_wrappers.trading_env.py` Relevant config keys: (none) Exercised flow: The agent’s portfolio value is tracked, and during market drops, the agent liquidates positions, increasing the cash weight. Why this code matters: The environment’s constraints or the agent’s inability to learn hedging strategies result in liquidation to cash during market downturns.

Hypothesis 9

Claim text: PPO’s higher transaction frequency indicates a momentum-chasing strategy compared to the value-oriented approach of TD3. Relevant code paths: `evaluation.runner.py` Relevant config keys: `ppo_params`, `td3_params` Exercised flow: The runner compares the transaction counts and performance metrics of PPO and TD3. Why this code matters: PPO’s policy updates encourage more frequent trading, which in this environment manifests as momentum chasing, whereas TD3’s deterministic policy is more stable.

Hypothesis 10

Claim text: The agents’ reliance on a small subset of tickers suggests overfitting to the training data’s specific sector leaders. Relevant code paths: `data.ticker_presets.py` Relevant config keys: (none) Exercised flow: The ticker list is resolved in `experiment_setup.py`, and the agent learns to trade only a

subset of these. Why this code matters: Agents are trained on a fixed set of tickers, and the lack of generalization leads them to focus only on the most prominent ones.

Hypothesis 11

Claim text: PPO exhibits significant ‘buy-bias’ and over-concentration compared to other agents. Relevant code paths: `envs.gym_wrappers.trading_env.py` Relevant config keys: `ppo_params` Exercised flow: The agent’s actions are processed in the trading environment, where buy orders are executed and tracked. Why this code matters: PPO’s policy optimization often leads to aggressive buying behavior, resulting in over-concentration in specific assets.

Hypothesis 12

Claim text: A2C and DDPG suffer from ‘transaction paralysis’ due to overly conservative reward functions. Relevant code paths: `envs.gym_wrappers.trading_env.py` Relevant config keys: (none) Exercised flow: The agent receives a negative reward for every trade, causing it to learn a policy that minimizes trading. Why this code matters: The reward function penalizes transaction costs heavily, leading the agents to avoid trading altogether.

Hypothesis 13

Claim text: TD3 demonstrates superior risk-aversion through higher ‘HOLD’ frequency. Relevant code paths: `envs.gym_wrappers.trading_env.py` Relevant config keys: `td3_params` Exercised flow: The agent’s action history shows a higher proportion of ‘HOLD’ actions, which is interpreted as risk-aversion. Why this code matters: TD3’s architecture, which includes target networks and delayed policy updates, leads to more stable and risk-averse behavior.

Hypothesis 14

Claim text: Agents are failing to adapt to synthetic regime shifts, as evidenced by consistent ticker preferences across the entire test period. Relevant code paths: `data.fetchers.experiment_data_loader.py` Relevant config keys: (none) Exercised flow: The agent’s ticker preferences are tracked throughout the test period, showing no change despite synthetic regime shifts. Why this code matters: The agents’ inability to adapt to regime shifts is due to the lack of explicit regime-aware features in the state space.

Hypothesis 15

Claim text: The high cash weight range (0% to 100%) indicates that agents are frequently liquidating portfolios due to ‘fear’ signals rather than strategic cash management. Relevant code paths: `envs.gym_wrappers.portfolio_env.py`

Relevant config keys: (none) Exercised flow: The portfolio environment tracks cash weight, which spikes during periods of high turbulence. Why this code matters: The agent’s liquidation behavior is a reaction to high volatility or turbulence, which is interpreted as ‘fear’.

Hypothesis 16

Claim text: DDPG and TD3 agents are overly conservative, favoring ‘HOLD’ actions due to high transaction costs or reward penalties in the environment. Relevant code paths: `envs.gym_wrappers.trading_env.py` Relevant config keys: `ddpg_params`, `td3_params` Exercised flow: The agent’s action history is analyzed, showing a high frequency of ‘HOLD’ actions. Why this code matters: The environment’s transaction costs act as a barrier to entry, causing DDPG and TD3 to favor ‘HOLD’ actions.

Hypothesis 17

Claim text: The agents’ trading behavior is regime-dependent, specifically failing to rotate out of assets during synthetic bear market segments. Relevant code paths: `envs.gym_wrappers.trading_env.py` Relevant config keys: (none) Exercised flow: The agent’s portfolio composition is tracked during bear market segments, showing no rotation. Why this code matters: The agents lack the capability to detect bear market regimes and rotate assets, leading to losses.

Hypothesis 18

Claim text: A2C exhibits ticker-specific momentum chasing, evidenced by concentrated buying in high-beta tickers during specific time windows. Relevant code paths: `envs.gym_wrappers.trading_env.py` Relevant config keys: `a2c_params` Exercised flow: The agent’s buy orders are correlated with high-beta ticker performance during specific time windows. Why this code matters: A2C’s policy optimization leads it to identify and exploit momentum in high-beta tickers.

Hypothesis 19

Claim text: The ‘Cash weight’ volatility is driven by the agents’ inability to find suitable entry points, resulting in prolonged periods of 100% cash. Relevant code paths: `envs.gym_wrappers.portfolio_env.py` Relevant config keys: (none) Exercised flow: The portfolio environment tracks cash weight, which remains at 100% for extended periods. Why this code matters: The agent’s inability to identify profitable entry points leads to a default to cash, causing high volatility in cash weight.